

Erdős Problem #695

EP-695

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $p_1 < p_2 < \dots$ be a sequence of primes such that $p_{i+1} \equiv 1 \pmod{p_i}$. Is it true that $\lim_k p_k^{1/k} = \infty$?
Does there exist such a sequence with $p_k \leq \exp(k(\log k)^{1+o(1)})$?